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Control device and manufacturing method

Abstract

The invention relates to a control device having a casing, control electronics located in the casing, and at least one electrical feedthrough. By means of the electrical feedthrough, a current is provided for a load which is located outside the casing of the control device. The electrical feedthrough is located on the casing. The casing is fluid-tight at least in part, and the electrical feedthrough is located in a fluid-tight region of the casing. The electrical feedthrough has a metal core and an insulator surrounding the metal core. The metal core has an end face. A first bond is provided between the end face of the metal core and the control electronics, the first bond being laser-bonded at the end face.

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Background/Summary

BACKGROUND

(1) The invention relates to a control device and a method for manufacturing a control device.

(2) From the prior art, it is known to seal control devices against gaseous or liquid media with control electronics located in the control device. To provide electrical currents or voltages outside of a sealed casing of the control device, electrical feedthroughs can be used. These feedthroughs usually consist of a round metal core, wherein the round metal core is overcast with glass in order to achieve electrical insulation of the metal core against the casing. The combination of glass and metal core leads to a seal against fluids, i.e., gases or liquids. To connect the metal core to the control electronics, guide elements can be welded or soldered or attached. During soldering or welding, it can occur that cracks are formed in the glass due to the high temperatures during the welding or soldering process, so that the electrical feedthrough is not tightly sealed against gases or liquids. Even with plug connections, mechanical impacts may occur on the insulator, which lead to leaks. Furthermore, a corresponding plug element must be located on the side of the control electronics, which makes the manufacturing method more complex.

SUMMARY

(3) One problem addressed by the invention is to provide a control device in which an electrical feedthrough is reliably connected to control electronics and, during this process, damage to an insulator of the electrical feedthrough, in particular containing glass, is unlikely. A further problem addressed by the invention is to specify a corresponding manufacturing method.

(4) A control device has a casing, control electronics located in the casing, and at least one electrical feedthrough. By means of the electrical feedthrough, current can be provided for a load which is located outside the casing of the control device. Preferably, the load is an electrically driven compressor, and the control device preferably controls the electrically driven compressor. Preferably, the load can represent a component through which fluid flows in operation, in particular a valve or a pump.

(5) The electrical feedthrough is located on the casing, and the casing is fluid-tight at least in part. The electrical feedthrough is located in a fluid-tight region of the casing. The electrical feedthrough has a metal core and an insulator surrounding the metal core. The metal core has an end face, wherein a first bond is provided between the end face of the metal core and the control electronics. The first bond is laser-bonded at the end face.

(6) With the insulator surrounded by the metal core, the metal core of the electrical feedthrough is insulated against the casing, so that the current for the load can be provided well outside the casing without generating a short circuit with the casing. The casing may be completely fluid-tightly closable and may have, for example, a first casing half and a second casing half, wherein a seal is provided between the casing halves. The control device can be installed, for example in a vehicle, such that fluids, i.e., gases or liquids, can reach the casing from the outside at least in the fluid-tight region of the casing, and a penetration of the fluids in the fluid-tight region of the casing is to be prevented. The end face of the metal core can be located within the casing such that the first bond provides an electrical connection between the metal core and the control electronics and, via this first bond, the current can be passed from the control electronics outwardly to the load through the electrical feedthrough.

(7) The insulator can contain glass and can be configured in the shape of a glass layer, for example. The glass forms an electrical and a fluid insulator. The insulator prevents fluid from flowing into the casing in the region of the electrical feedthrough. Specifically, when the load is configured as a

compressor, the fluid within a load casing of the load can have a high pressure. If a part of the casing also simultaneously forms the load casing, then the fluid acts with the pressure directly on the elements in the region of the electrical feedthrough. The insulator can also contain a rubber or a ceramic. Combinations of the aforementioned materials are also possible.
(8) The first bond can consist of a bond wire or a bond ribbon. Due to the fact that the first bond is laser-bonded at the end face, the end face is electrically connected to the first bond, wherein, due to the laser-bonding process, there is a connection region at the transition between the end face and the first bond, wherein the connection region consists of melted and resolidified material of the first bond as well as the metal core.
(9) A method for manufacturing a control device comprises the following steps: providing a casing, wherein the casing has at least one electrical feedthrough, wherein the casing is fluid-tight at least in part, wherein the electrical feedthrough is located in a fluid-tight region of the casing, wherein the electrical feedthrough has a metal core and an insulator surrounding the metal core, wherein the metal core has an end face; placing control electronics in the casing; placing a first bond between the end face of the metal core and the control electronics; laser-bonding the first bond at the end face.
(10) The laser-bonding of the first bond at the end face can occur in such a way that the first bond is brought into mechanical contact with the end face, and the first bond and the underlying end face are subsequently heated and partially melted with a focused laser beam. After the laser is switched off, a connection region is then formed between the first bond and the end face, wherein initially melted and subsequently resolidified material of both the first bond and the metal core is located in the connection region. This results in an electrically conductive and mechanically stable connection between the metal core and the first bond.
(11) For example, the laser radiation used for the laser-bonding can be provided with a fiber laser having about 1000 Watts of power and 1070 nm emission wavelength. However, other types of lasers, laser powers, and wavelengths are contemplated. The laser beam can optionally be guided in a circularly oscillating manner in order to achieve a greater impact surface of the laser beam on the first bond and thus to achieve a large connection surface and a low transition resistance. The placement of the first bond and the laser-bonding can be carried out in a joint step, if necessary. For this purpose, commercial systems are available in which the first bond and the laser-bonding are carried out in one step, which have in particular the above-mentioned properties of the laser.
(12) By the placement of the first bond and the laser-bonding the first bond with the metal core, several advantages for the control device and the manufacturing method are achieved. First, a probability of damage to the insulator, in particular a glass or a ceramic of the insulator, is reduced compared to the creation of a connection by welding or soldering process, because the temperatures occurring in the laser-bonding are significantly lower. Furthermore, the first bond can be movable to some extent, thereby compensating for manufacturing tolerances, for example of the electrical feedthrough, as well as for different thermal expansions of the electrical feedthrough and control electronics.
(13) In one embodiment of the control device, the control electronics are located on a printed circuit board and the first bond is laser-bonded at the printed circuit board. For the manufacturing method, the first bond is thus placed on the printed circuit board and also laser-bonded at the printed circuit board analogously to the procedure of laser-bonding between the first bond and the end face of the metal core. Laser-bonding is a joining method suitable for thin workpieces, in particular bonds, preferably bond wire or bond ribbon, in the range of approximately 100 μm, so that the first bond can optionally be laser-bonded directly to the printed circuit board. This allows for a particularly simple manufacturing method.
(14) In one embodiment of the method, a buffer plate is located on the printed circuit board. The first bond is laser-bonded at the buffer plate. The buffer plate can be a cuboid body made of a metal that is attached to the printed circuit board, for example soldered, by conventional methods. In the

manufacturing method, for example, the buffer plate can be mounted and soldered onto the printed circuit board, or the buffer plate is already located on the printed circuit board of the control electronics. In both cases, the buffer plate can be configured as an SMD part and can be connected to the control electronics via SMD soldering. The buffer plate offers advantages in laser-bonding due to the fact that, firstly, a thicker metal layer (the buffer plate itself) is available for the laser-bonding process due to the buffer plate and, secondly, the buffer plate allows for an easier dissipation of the process heat generated during the laser-bonding.

(15) In one embodiment, the printed circuit board has a thermally conductive structure in the region of the first bond. The thermally conductive structure is thermally connected to the casing and electrically insulated against the casing. In the manufacturing method, the thermally conductive structure can already be part of the printed circuit board of the control electronics. The thermally conductive structure can be formed in the shape of vias, in particular copper vias, so that a heat can be conducted through the vias. A thermal compound, for example based on silicon, can be placed between the printed circuit board and the casing. The thermal compound can have a good thermal conductivity and a poor electrical conductivity, such that, by means of the thermal compound, the thermal structure is electrically insulated against the casing without impeding the conduction of heat from the thermally conductive structure to the casing. In the manufacturing method, it can then be provided that the thermal compound is applied at this point in the casing before the control electronics are placed.

(16) The metal core of the electrical feedthrough can be configured so as to be substantially round. This means that the metal core is configured so as to be round, in particular in the region of the insulator, and the insulator surrounds the metal core as a round sheath layer. This is easier to achieve in terms of process technology than other cross-sections of the metal core.

(17) In one embodiment of the control device, the end face of the metal core has a pedestal protruding over the metal core. This pedestal can be formed integrally or in two pieces with the metal core. The metal core and the pedestal can thus be manufactured from one workpiece or manufactured from two different workpieces and connected to one another. In this case, the end face of the metal core is the end face of the pedestal. As a result, an application surface of the electrical feedthrough can be increased and, if applicable, laser-bonding of the first bond at the end face can be simplified.

(18) In one embodiment of the control device, the electrical feedthrough is configured such that the current supplied for the load is greater than 1 ampere, in particular greater than 10 amperes, and preferably in the range of 100 amperes. In this case, at least one bond, in particular at least one bond wire and/or at least one bond ribbon, is in any case to be provided for higher currents, in particular in the range of 10 or more amperes, because bond wires are no longer suitable for these high currents. For particularly high currents in the range of 100 amperes, it can additionally be provided that the first bond has more than a single bond ribbon or a single bond wire, in particular two or more. This is particularly simple to carry out when the buffer plate is used on the printed circuit board, because the buffer plate then provides a sufficiently large surface area for the mounting of the first bonds. In this embodiment, if applicable, the use of the pedestal of the metal core is also advantageous in order to increase an existing surface area in the region of the electrical feedthrough.

(19) In one embodiment of the control device, a current supply is passed through the casing. The control electronics are connected to the current supply by means of a second bond, wherein the second bond is laser-bonded at the current supply and at the control electronics. The current supply can also be configured in the form of electrical feedthroughs, but also in the form of conventional plugs, depending on whether the casing is designed so as to be fluid-tight in the region of the current supply or not. It can also be provided that a buffer plate on a circuit board of the control electronics is provided in the region of the current supply in order to utilize the advantages of the buffer plate in the region of the current supply as described in connection with the first bond. In the

manufacturing method, the second bond is placed at the current supply and the control electronics and is also laser-bonded as described for the first bond. In particular, it is advantageous that the laser-bonding of the first bond and the laser-bonding of the second bond can be done in one work step, i.e., in particular within one machine.

(20) Furthermore, signal plugs can additionally be provided on the casing, with which the control electronics can receive control signals from the outside or can input measured values from sensors. These signal plugs can also be laser-bonded at the control electronics in order to also enable the electrical connection of the signal plugs with the control electronics in one work step.

(21) In one embodiment of the control device, the metal core consists of copper or iron. Copper is well-suited as a material for laser-bonding processes, wherein the metal core can contain, for example, pure copper having at least 99.99 percent purity, but also certain copper alloys. Such a copper alloy can have, for example, between 0.7 and 1.4 percent by weight chromium and low amounts of zirconium in addition to the copper. The rest of the alloy is copper. If the metal core consists of iron, in particular iron-nickel alloys can be provided, for example having between 45 and 55 percent by mass iron, between 45 and 55 percent by mass nickel, and up to 0.5 percent by mass silicon. These materials are also suitable for the laser-bonding process. The selection is made based on the thermal expansion and the electrical conductivity. The thermal expansion must be similar to the thermal expansion of the insulator and/or the metal plate. The aforementioned iron-nickel alloy has these advantageous properties. In one embodiment of the control device, the first bond consists of copper or aluminum. In particular, copper ribbons having a cross-section of 2 mm to 0.2 mm or 2 mm to 0.3 mm, or cross-sections of a similar magnitude, or aluminum ribbons of the same magnitude, are well suited for creating a laser bond with the metal core. If a second bond to the current supply is provided, it can consist of the same material as the first bond.

(22) In one embodiment of the control device, the control electronics has a motor control. The motor control is configured so as to provide three-phase actuation of the load configured as a motor. To enable this, at least one electrical feedthrough is provided for each motor phase, each of which is connected to the control electronics by means of first bonds. Preferably, more or less can also be formed, in particular two, four, or six electrical feedthroughs. Thus, three electrical feedthroughs are located in the casing, each connected to the control electronics by means of a first bond, and each being laser-bonded at the metal core of the associated electrical feedthrough as well as the control electronics. The three electrical feedthroughs can serve to provide one of the three phases, also designated with the letters UVW, respectively. Instead of the three-phase actuation, a different multi-phase actuation can also be provided, wherein a separate electrical feedthrough can be provided for each of the phases of the multi-phase actuation. In particular, electrically driven compressors that are electrically commutated have three phases and thus electrical feedthroughs.

(23) In particular, a motor can be provided as the load, wherein it can be provided that the motor drives a compressor, a fan, a pump, a window regulator, or an expansion valve for a refrigerant. The motor and the compressor can, for example, be exposed to a gas or a liquid that must not enter the control electronics, so that an advantageous configuration can be achieved by the aforementioned sealing of the casing. Preferably, a compressor or valve has a high-pressure fluid. This fluid must not enter the casing. The same applies when an expansion valve for a refrigerant can be driven by means of the control device, because the refrigerant could possibly also be harmful to the control electronics. Also when used in a hydrogen system, the hydrogen must not enter the casing. The control device according to the invention is further possible as a transmission control, wherein elements to be driven can be located outside the casing within the transmission and thus can be exposed to the transmission oil, wherein the transmission oil also must not enter the control device or the casing of the control device. The control device can further be advantageously used with the inverter of an electric vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the invention are explained with reference to the following drawings. In the schematic drawings, the following are shown:

(2) FIG. **1** a cross-section through a control device with a load;

(3) FIG. **2** an enlargement of a connection between a metal core and a first bond;

(4) FIG. **3** a cross-section through a further control device;

(5) FIG. **4** a flowchart of a manufacturing method;

(6) FIG. **5** an isometric view of a control device;

(7) FIG. **6** an isometric view of a compressor; and

(8) FIG. **7** a feedthrough component.

DETAILED DESCRIPTION

(9) FIG. **1** shows a cross-section through a control device **100** having a casing **110** and control electronics **120** located in the casing **110**. Adjacent to the casing **110**, a load **200** is located in a load casing **201** having a load casing interior **202**. The load is in particular an electrically driven compressor. In a compressor, there is a fluid, in particular gaseous, within the load casing interior **202** and the pressure of the fluid can be greatly increased in relation to the pressure within the casing **110**. The casing **110** further has an electrical feedthrough **130**, wherein current can be provided to the load **200** located outside of the casing **110** by means of the electrical feedthrough **130**. The casing **110** is fluid-tight at least in part, and the electrical feedthrough **130** is located in a fluid-tight region **111** of the casing **110**. Thus, in the fluid-tight region **111**, the casing **110** is fluid-tight against fluids located within the load casing interior **202**, such as liquids or gases.

(10) The electrical feedthrough **130** has a metal core **131** having an end face **132**, wherein the end face **132** is located inside the casing **110**. Preferably, the metal core is cylindrical. The electrical feedthrough **130** further has an insulator **133** surrounding the metal core **131**, wherein electrical insulation of the metal core **131** against the casing **110** is achieved by the insulator **133**. Preferably, the insulator **133** coaxially surrounds the metal core. A first bond **140** is provided between the control electronics **120** and the end face **132** of the metal core **131**. The first bond **140** can be configured as a bond wire or bond ribbon and serves to guide a current from the control electronics **120** to the feedthrough **130** and further on to the load **200**. The first bond **140** is laser-bonded at the end face **132**. This means that the material of the first bond **140** is placed on the end face **132** and subsequently illuminated by a laser. By the laser irradiation, a connection region is created by interconnecting the metal core **131** and the first bond **140**. It is advantageous that the bond does not damage the insulator **133**. Such a damage can lower the fluid tightness, which can allow fluid to enter the casing **110**. Also, only a low force and heat input is necessary in the manufacturing of the bond.

(11) The insulator **133** can consist of glass, rubber, or ceramic, or a combination of these materials. Advantageously, the insulator **133** is formed from glass. The glass causes an electrical insulation and an insulation against the ingress of fluid. The insulator **133** can in particular be configured as a glass layer or a ceramic layer.

(12) FIG. **2** shows an enlarged section of the connection between the first bond **140** and the metal core **131**. A connection region **141** extends from the metal core **131** to the first bond **140**. During the laser-bonding, material of the metal core **131** and the first bond **140** is melted by the energy of the radiated laser and solidified to the connection region **141** upon deactivation of the laser, wherein material of both the metal core **131** and the first bond **140** is located in the connection region **141**. To illustrate this, the connection region **141** is shown by a dashed line. Preferably, the laser-bonding allows a bonding region that protrudes into the bond **140** as much as into the metal core. The laser-bonding allows a well-dosable and targeted amount of energy to be used.

(13) FIG. **3** shows a cross-section through a further exemplary embodiment of a control device **100**, which basically corresponds to the control device **100** of FIG. **1**, but has further optional features. The individual additional features described below are also individually additionally implementable in the control device **100** of FIG. **1**. Various such exemplary embodiments, optionally also implementable in the control device **100** of FIG. **1**, are described below.
(14) The control electronics **120** are located on a printed circuit board **121** in one exemplary embodiment. A printed circuit board is understood to mean a circuit carrier, in particular a carrier of electronic components. The printed circuit board consists of epoxy resin, paper, fiber-reinforced plastic, ceramic substarts (DBC, LTCC, LCTC), or insulated metal (IMS). Accordingly, the printed circuit board is configured as a DBC, LTCC, LCTC, IMS printed circuit board. The first bond **140** is laser-bonded at the printed circuit board **121**. This can be done, for example, by laser-bonding the first bond **140** on a conductor path of the printed circuit board **121**. Thus, the first bond **140** can be laser-bonded directly to the printed circuit board **121**.
(15) In one exemplary embodiment, as shown in FIG. **3**, a buffer plate **122** is located on the printed circuit board **121** and the first bond **140** is laser-bonded at the buffer plate **122**. In particular, the buffer plate **122** provides a greater material thickness, so that the laser-bonding process can be easier to configure with a connection region analogous to FIG. **2**.
(16) It is also shown in FIG. **3**, as a further exemplary embodiment, that the printed circuit board **121** has a thermally conductive structure **123** in the region of the first bond **140**. The thermally conductive structure **123** can consist of vias guided through the printed circuit board **121**. The thermally conductive structure **123** is thermally conductively connected to the casing **110** and electrically insulated against the casing **110**. This can be done, for example, by a thermally but not electrically conductive thermal compound **124** being provided between the printed circuit board **121** and the casing **110** in the region of the thermally conductive structure **123**. The thermal compound **124** can contain silicon.
(17) In one exemplary embodiment, as shown in FIG. **3**, the casing **110** is produced of a bottom part **112** having the fluid-tight region **111** with the electrical feedthrough **130** and a lid **113**. A seal **114** is located between the lid **113** and the bottom part **112**, such that the entire casing **110** can be fluid-tight.
(18) In one exemplary embodiment, the electrical feedthrough **130** is configured as a feedthrough device **135** with a metal plate **134**. The metal core **131** with the insulator **133** is located on the metal plate **134** and attached to the fluid-tight region **111** of the casing **110** with a seal **114**. This allows the feedthrough device **135** to be optionally removably positioned on the casing **110** and thus to replace defective electrical feedthroughs **130**.
(19) The metal plate **134** for each electrical feedthrough **130** has in particular a recess, in particular a cylindrical, continuous recess, preferably a bore. The electrical feedthrough **130** is guided through the recess. The recess is in particular configured as a continuous recess. If no metal plate is configured, a corresponding recess is configured in the casing. By way of example, the metal plate **134** in FIG. **3** forms part of the casing **110** of the control device **100** and part of the load casing **201** of the load **200** at the same time. The metal plate in turn closes a recess in the casing **110**. Preferably, it closes a recess between the housings **110** of the control device and the load casing **201**. Advantageously, the metal plate **134** can be manufactured with the electrical feedthroughs **130** and the insulator **133** and then inserted as a part into the recess provided for this purpose. In particular, the metal plate is attached to one or both housings by means of welding, soldering, gluing, screwing, riveting. The metal plate is sealed via a further insulator, in particular a sealing ring or also glass.
(20) An insulator **133** is formed around the metal core **131** of the electrical feedthrough **130**. The insulator is particularly coaxially formed around the metal core **131**. The insulator is at least in part liquefied after being positioned within the recess. As a result, the insulator closes the open region around the metal core **131**. The insulator **133** closes openings and open regions between the recess

and the metal core **131**.

(21) In one exemplary embodiment, the casing **110** additionally has a current supply **115** guided through the casing **110**. The current supply **115** is configured with a plate, in particular a metal plate, a metal core, and an insulator **133**, analogously to the electrical feedthrough **130**. This is particularly advantageous in combination with the already described exemplary embodiment with the completely fluid-tight casing **110**, because, in this case, a seal against fluids is also possible in the region of the current supply **115**. If the casing **110** is only fluid-tight in the fluid-tight region **111**, a conventional, non-fluid-tight plug can optionally be installed instead of the current supply **115** shown in FIG. **3**. The control electronics **120** are connected to the current supply **115** by means of a second bond **150**. The second bond **150** is laser-bonded at the current supply **115**. Furthermore, it can likewise be provided that the second bond **150** is also laser-bonded at the control electronics **120**, in particular the printed circuit board **121**. Also optionally, an additional buffer plate **122** can also be provided here.

(22) The printed circuit board **121** of the control electronics **120** additionally has electronic components **125** that can serve to control the load **200**.

(23) In the control devices **100** shown in FIGS. **1** to **3**, it can be provided that the metal core **131** consists of copper or iron. In particular, the metal core **131** can be formed from pure copper of at least 99.99 percent copper content or from a copper alloy having 0.7 to 1.4 percent chromium and a low proportion of zirconium. The alloy must have a thermal expansion similar to the insulator **133** as well as a high conductivity. Different thermal expansions lead to shifting of the insulator. Low conductivity leads to an increase in the cross-section. The alloys listed herein meet the specifications in an inventive manner. Furthermore, the metal core **131** can be configured as an iron-nickel alloy, in particular having an iron content of between 45 and 55 percent and a nickel content of between 45 and 55 percent as well as up to 0.5 percent silicon as alloy constituents. For example, an alloy of 52 percent nickel and 48 percent iron can be provided without silicon. Alternatively, an alloy having between 50.8 and 51.2 percent nickel, between 48.8 and 49.2 percent iron, and up to 0.3 percent silicon can be provided. The percentages can refer to mass percents.

(24) The buffer plate **122** can be constructed of the same materials as the metal core **131**, and in particular can consist of copper.

(25) The first bond **140** can consist of copper or aluminum. The second bond **150** can also consist of copper or aluminum.

(26) FIG. **4** shows a flowchart **300** of the method for manufacturing a control device **100** as shown in FIG. **1** or **3**. In a first method step **301**, the casing **110** is provided. The casing **110** has the electrical feedthrough **130** and is in part fluid-tight, wherein the electrical feedthrough **130** located in a fluid-tight region **111** of the casing **110**. The electrical feedthrough **130** has a metal core **131** and an insulator **133** surrounding the metal core **131**, wherein the metal core **131** further has an end face **132**. In the first method step **301**, it can be provided that the electrical feedthrough **130**, configured as a feedthrough device **135**, is first inserted into the casing **110**. The feedthrough device **135** has the electrical feedthroughs **130** and the insulator **133**. The feedthrough device **135** has a recess, in particular a cylindrical, continuous bore, into which an electrical feedthrough **130** is inserted. The insulator **133** is melted, thereby fluidly sealing the region between the metal core of the electrical feedthrough **130** and the metal plate.

(27) If the casing **110** is configured with a bottom part **112** and a lid **113** as shown in FIG. **3**, it can be provided that only the bottom part **112** of the casing **110** is initially provided, and the lid **113** is only attached after further method steps. In a second method step **302**, the control electronics **120** are placed in the casing **110** and fixed or screwed, for example. In a third method step **303**, the first bond **140** is located between the end face **132** and the control electronics **120**.

(28) In a fourth method step **304**, the first bond **140** is laser-bonded at the end face **132**, thereby forming the connection region **141** shown in FIG. **2**. In an optional fifth method step **305**, the first bond **140** can also be laser-bonded at a printed circuit board **121** or buffer plate **122**.

(29) In an optional sixth method step **306**, the second bond **150** of FIG. **3** can be located between the control electronics **120** and the current supply **115** and can also be laser-bonded, wherein additionally a laser-bonding of the second bond **150** is also possible at the control electronics **120** and the printed circuit board **121**, respectively.
(30) FIG. **5** shows an isometric view of a control device **100**, wherein the casing **110** has a bottom part **112** as in FIG. **3** and, so that the components located within the casing **110** are visible, the lid **113** shown in FIG. **3** is omitted. Of course, a lid **113** can be placed on the bottom part **112** of the casing **110** analogously to FIG. **3**. The control electronics **120** in turn have a printed circuit board **121**. The control electronics **120** have a motor control. The motor control is configured so as to perform a three-phase actuation of the load **200** configured as a motor. To enable this, three electrical feedthroughs **130** are provided, each having an end face **132** and each connected to two first bonds **140** with a buffer plate **122** on the printed circuit board **121**. A laser bond is configured between the first bonds **140** and the feedthroughs **130**, as well as between the first bonds **140** and the buffer plate **122**, respectively, as already described. For each electrical feedthrough **130**, two first bonds **140** are provided in order to be able to increase a total current flowing through the electrical feedthroughs **130**.
(31) The first bonds **140** are each configured as bond ribbons in the exemplary embodiment shown in FIG. **5**. For example, such bond ribbons can have a cross-section of 2 mm to 0.3 μm and are thus also suitable for high currents. In particular, it can be provided that currents greater than 1 ampere flow through the electrical feedthroughs **130** (for example, also in the exemplary embodiments of FIGS. **1** and **3**). For motor controls, it can in particular also be provided that the currents are greater than 10 amperes or even in the range of 100 amperes. In particular, two first bonds **140** per electrical feedthrough **130** are provided in order to enable 100 amperes through the electrical feedthroughs **130**. Below the buffer plates **122**, the thermally conductive structures already described in connection with FIG. **3** are provided with a good thermal connection to the casing **110** and electrical insulation against the casing **110**. With the currents typically flowing for motor control in the range of 100 amperes, it can be achieved that, in operation, the first bonds **140** can heat up to a maximum of 200° C. and the buffer plates **122** can heat up to a maximum of 100° C. and thus the required currents can be provided without destruction of the components.
(32) The bottom part **112** of the casing **110** additionally has two poles of a current supply **115**, each having a metal plate **116** at its upper end. The metal plates **116** are each connected to two second bonds **150** with buffer plates **122** of the printed circuit board **121**, wherein the second bonds **150** are laser-bonded at the metal plates **116** and the buffer plates **122**, respectively. Here, too, two second bonds **150** are provided in order to be able to accommodate the present currents.
(33) FIG. **6** shows an isometric view of the control device **100** of FIG. **5** mounted on a load **200**. The load **200**, not shown due to the load casing **201**, is configured as an electric motor and is used in order to drive a compressor **205**. The load is an electric motor that drives the compressor **205**. The gas flowing through the compressor **205** enters the load casing **201** and is shielded from the interior of the casing **110** by the electrical feedthroughs **130** and the fluid-tight region **111** of the casing **110**. Thus, gases compressed with the compressor **205** that are harmful for the control electronics **120** cannot reach the control electronics **120**. The lid **113** is also not shown in FIG. **6**, as in FIG. **5**.
(34) Instead of the electric motor and compressor **205**, as shown in FIGS. **5** and **6**, the load **200** can also have a valve, for example an expansion valve for a refrigerant. Also, the load can represent part of a fuel cell, for example a hydrogen valve or a hydrogen pump. Furthermore, an electric motor can be provided, which is actuated by the control electronics **120** and drives a fan or a coolant pump, wherein the gas or coolant moved by the fan also cannot pass to the control electronics **120**, and therefore the fluid-tight region **111** must also be provided. Alternative consumers can also be, for example, actuators in an automatic transmission, a current window motor, a pump, for example a gasoline pump, or an inverter of an electric vehicle. These

components can also be exposed to fluids that must not be passed to the control electronics **120**.
(35) FIG. **7** shows a feedthrough device **135**, as can be used in control device **100** of FIGS. **5** and **6**. The feedthrough device **135** has three electrical feedthroughs **130**, each having a metal core **131** and an insulator **133** inserted into a metal plate **134**. The end face **132** has a pedestal **136** protruding over the metal core **131**. This is particularly advantageous when the metal core **131** has a diameter that optionally does not allow for two or more first bonds **140** to be placed at the end face **132**. With the pedestal **136** protruding over the metal core **131**, a surface of the end face **132** is increased. The pedestal **136** can be formed integrally with the metal core **131** or can be welded to the metal core **131** or connected to the metal core **131**. However, in each case, the end face **132** refers to the end face of the overall component consisting of the metal core **131** and the pedestal **136**.
(36) In addition to the electrical feedthroughs **130** and current supplies **115** shown in the figures, it can be provided that signal plugs are located on the casing **110**, which are also connected to the control electronics **120** within the casing **110** by means of a laser-bonding process. This enables an overall connection of the control electronics with all relevant components in one work step by means of the laser-bonding process. In particular, the electrical feedthroughs **130** having a metal core **131** and insulator **133** are not susceptible to destruction during the laser-bonding process, such that even after laser-bonding, the electrical feedthroughs **130** are fluid-tight. This enables manufacturing advantages compared to welding, soldering, and plug methods.
(37) Although the invention has been described in more detail using preferred exemplary embodiments, the invention is not limited by the disclosed examples, and other variations can be derived from this by a person skilled in the art without departing from the scope of protection of the invention.

Claims

1. A control device (**100**) comprising a casing (**110**), control electronics (**120**) located in the casing (**110**), and at least one electrical feedthrough (**130**), wherein, by the electrical feedthrough (**130**), a current is provided for a load (**200**) which is located outside the casing (**110**) of the control device (**100**), wherein the electrical feedthrough (**130**) is located on the casing (**110**), wherein the casing (**110**) is fluid-tight at least in part, wherein the electrical feedthrough (**130**) is located in a fluid-tight region (**111**) of the casing (**110**), wherein the electrical feedthrough (**130**) has a metal core (**131**) and an insulator (**133**) surrounding the metal core (**131**), wherein the metal core (**131**) has an end face (**132**), wherein a first bond (**140**) is provided between the end face (**132**) of the metal core (**131**) and the control electronics (**120**), wherein the first bond (**140**) is laser-bonded at the end face (**132**), wherein the control electronics (**120**) is located on a printed circuit board (**121**) and the first bond (**140**) is located on the printed circuit board (**121**), wherein a buffer plate (**122**) is located on the printed circuit board (**121**) and the first bond (**140**) is laser-bonded at the buffer plate (**122**), wherein the printed circuit board (**121**) has a thermally conductive structure (**123**) in a region of the first bond (**140**), wherein the thermally conductive structure (**123**) is thermally conductively connected to the casing (**110**), wherein the thermally conductive structure (**123**) is electrically insulated against the casing (**110**).
2. The control device (**100**) according to claim 1, wherein the end face (**132**) has a pedestal (**136**) protruding over the metal core (**131**).
3. The control device (**100**) according to claim 1, wherein a current supply (**115**) is passed through the casing (**110**) and wherein the control electronics (**120**) is connected to the current supply (**115**) by a second bond (**150**), wherein the second bond (**150**) is laser-bonded at the current supply (**115**) and at the control electronics (**120**).
4. The control device (**100**) according to claim 1, wherein the metal core (**131**) includes copper or iron.

5. The control device (**100**) according to claim 1, wherein the first bond (**140**) includes copper or aluminum.
6. The control device (**100**) according to claim 1, wherein the control electronics (**120**) has a motor control, wherein end faces (**132**) of three electrical feedthroughs (**130**) are connected to the control electronics (**120**) by first bonds (**140**), and wherein the motor control is configured to carry out a three-phase actuation of the load (**200**) via the three electrical feedthroughs (**130**).
7. The control device (**100**) according to claim 1, wherein the insulator (**133**) is formed from glass.
8. The control device (**100**) according to claim 1, wherein the load (**200**) is configured as a compressor having a compressor casing, and wherein there is increased pressure within the compressor.
9. A method for manufacturing a control device (**100**), wherein the method comprises the following steps: providing (**301**) a casing (**110**), wherein the casing (**110**) has at least one electrical feedthrough (**130**), wherein the casing (**110**) is fluid-tight at least in part, wherein the electrical feedthrough (**130**) is located in a fluid-tight region (**111**) of the casing (**110**), wherein the electrical feedthrough (**130**) has a metal core (**131**) and an insulator (**133**) surrounding the metal core (**131**), wherein the metal core (**131**) has an end face (**132**); placing (**302**) control electronics (**120**) in the casing (**110**) and positioning the control electronics (**120**) on a printed circuit board (**121**); positioning a buffer plate (**122**) on the printed circuit board (**121**); placing (**303**) a first bond (**140**) between the end face (**132**) of the metal core (**131**) and the control electronics (**120**), wherein the first bond (**140**) is positioned on the printed circuit board (**121**), wherein the printed circuit board (**121**) has a thermally conductive structure (**123**) in a region of the first bond (**140**), wherein the thermally conductive structure (**123**) is thermally conductively connected to the casing (**110**), wherein the thermally conductive structure (**123**) is electrically insulated against the casing (**110**); and laser-bonding (**304**) the first bond (**140**) at the end face (**132**) and the buffer plate (**122**).
10. The control device (**100**) according to claim 1, wherein the insulator (**133**) seals the casing (**110**) against an ingress of fluid in a region of the electrical feedthrough (**130**).